

Motor Claims Delay and Payment Control

A P&C insurance business analysis case study

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What this case study proposes

A claim should not disappear into a queue because a policy number is wrong or a document is missing. This case study proposes a claims workflow that identifies the issue, gives it an owner, and shows what needs to happen next. It also checks payment requests against approved amounts before money is released.

The design brings together requirements, process mapping, source to target mapping, data lineage, SQL controls, dashboard specifications, and UAT. A recruiter can follow one requirement from the business problem through to the acceptance test.

Project status

This is an independent portfolio simulation for Harbor Motor Insurance, a fictional insurer. The processes, service levels, amounts, examples, and business targets are design assumptions. No insurer data was used. The document does not claim a live implementation, client delivery, Guidewire configuration experience, or measured business improvement. SQL examples were checked against a small synthetic fixture; the workflow and dashboard are proposed designs.

Reading guide

Pages cover the business case and scope, insurance concepts, current and proposed processes, requirements, exception rules, data structures, STTM, integration controls, SQL examples, reporting, UAT, delivery planning, and interview preparation.

Business case and project scope

Harbor Motor Insurance wants to reduce avoidable delays in personal motor physical damage claims. In the assumed current process, intake staff check information manually, assessors receive incomplete cases, and finance receives payment requests through a separate channel. Managers use spreadsheets to chase work. These are scenario assumptions rather than findings from an actual insurer.

Business objectives

Objective	Proposed success measure
Find delayed work sooner	Every open exception has a named queue, creation time, due date, and current status.
Reduce repeated handling	Measure touches per claim and repeat requests for the same document.
Improve payment control	Prevent a new request from exceeding the remaining approved exposure amount.
Make reporting dependable	Reconcile source counts and signed financial totals for each load.

Included in the first release

Personal motor collision and non-collision physical damage claims; FNOL; policy and coverage verification; document checks; assessor assignment; assessment; settlement approval; payment validation; closure; reopened claims; and an exception dashboard. Each claim can contain several exposures, but the worked financial example uses one exposure.

Boundaries

Pricing, underwriting decisions, policy issuance, bodily injury adjudication, litigation, regulatory reporting, automated fraud decisions, and payment execution are outside this release. Recovery records are captured for reporting, but a full subrogation workflow is a later phase. Third party liability is included in the glossary to show the wider domain, rather than built into the first release.

Assumptions and dependencies

One reporting currency, USD, is used throughout. The scenario uses US personal auto terminology without applying any specific state rules. Policy terms govern actual coverage. All timestamps are stored in UTC. Proposed SLAs use elapsed calendar hours. Product owners must approve the rule set, required documents, authority limits, and source extracts before a real implementation.

Insurance concepts and stakeholders

Term	Meaning in this case study
Property and casualty	Insurance addressing property losses and liability risks.
Collision	Damage to the insured vehicle from a collision, subject to policy terms.
Comprehensive	Non-collision vehicle damage, such as theft or flood, subject to policy terms.
Third party liability	Liability for injury or property damage suffered by others.
FNOL	First notice of loss, when the insurer first receives the loss report.
Exposure	A specific potential payment obligation within a claim, linked here to a coverage.
Deductible	The amount borne by the insured under applicable coverage terms.
Reserve	An estimate of future claim payments; it is not settlement approval.
Settlement approval	Authorized amount available for payment for an exposure.
Recovery and subrogation	Recovery is money received against claim costs; subrogation concerns pursuing a responsible party.
Endorsement	A change to policy terms with its own effective period.

Terminology is informed by the Insurance Information Institute and Guidewire documentation. The proposed data model is simplified and is not the Guidewire schema. Sources are listed at the end.

Stakeholder	Decision or responsibility
Claims operations manager	Owns workflow, queues, capacity, and service levels.
Claims handler and assessor	Checks documents, investigates loss, records assessment.
Coverage specialist	Reviews policy versions, exclusions, and unresolved coverage issues.
Finance approver	Approves payment authority and reconciles transactions.
Business analyst	Documents requirements, maps processes and data, coordinates acceptance.
Data engineer and BI analyst	Build ingestion, validation, reporting model, and measures.
QA and information security	Verify behavior, permissions, audit records, and data protection.

Current process and proposed process

Assumed current process

The customer reports a loss. Intake records the claim and asks another team to check the policy. Documents are collected through email. An assessor reviews the damage when the file appears complete. A handler prepares settlement approval, and finance checks a separate payment request. Delays are followed up manually. There is no consistent exception owner across these steps.

Proposed process

Register the claim immediately, retain the reported policy reference, and run intake checks. Verify the policy version and relevant coverage at the time of loss. Create required-document tasks and route review exceptions. Once ready, assign an assessor. After assessment, record an approved exposure settlement and run payment controls. Reconcile the posted payment before closure.

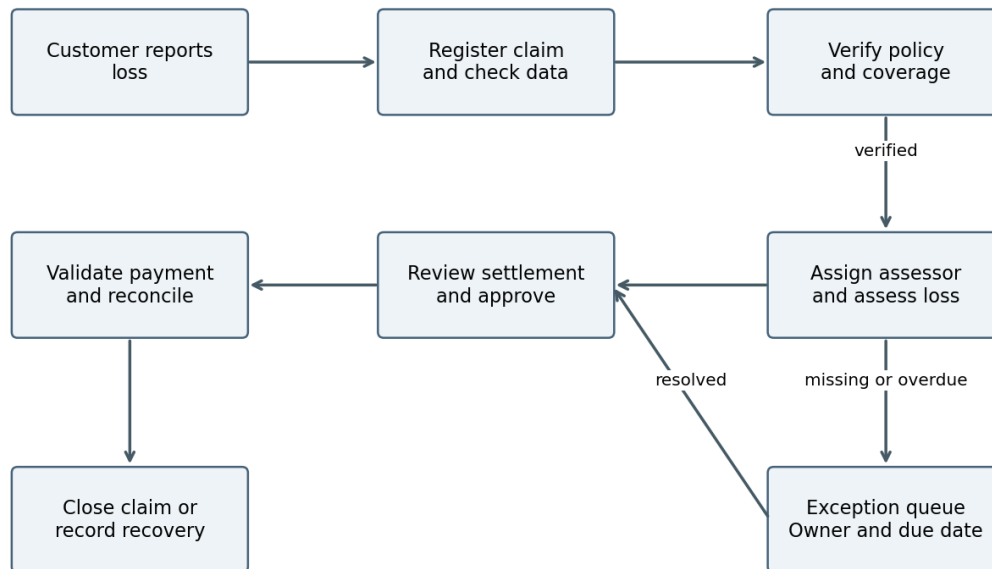


Figure 1 Proposed claim workflow. Coverage, approval, and payment failures also enter the exception queue. Each issue returns to its original stage after resolution.

A suspect claim remains available for review. A flag does not deny coverage or prove fraud. Payment remains blocked until required approvals and holds are resolved. Closing a claim is separate from closing its exposures and any recovery tasks.

A worked claim example

Claim CLM1001 concerns collision damage reported on 10 September 2026. The loss occurred on 9 September. The intake policy reference contains a typo. The following journey illustrates the proposed behavior, rather than a real customer incident.

Step	Record and action
1 Intake	Save claim with the original reference and create E01 for unmatched policy. Assign the coverage queue.
2 Verification	Handler corrects the reference with supporting evidence. Save both old and new values. Link the policy version effective at loss.
3 Documents	The product rule requires loss photos and a repair estimate. The estimate is absent, so E03 remains open.
4 Assessment	After documents are accepted, the assessor records an eligible repair cost of USD 4,000.
5 Settlement	For this simplified example only, a USD 500 deductible produces an approved amount of USD 3,500. Record the approval version and approver.
6 Payments	USD 2,000 has posted. A new USD 1,800 request exceeds the USD 1,500 remaining authority, so E06 blocks it.
7 Correction	Finance changes the request to USD 1,500. Validate again inside the payment transaction. Total posted becomes USD 3,500.
8 Closure	Reconcile the transaction reference, resolve open holds, close the exposure, then close the claim.

Why the example matters

It connects a customer problem to operational ownership, a coverage check, document readiness, a financial control, and reporting. The payment rule considers cumulative payments rather than comparing each payment individually with the approval.

Coverage decision boundaries

The example assumes the vehicle, driver, loss cause, territory, and policy terms are eligible. A real deductible calculation may also depend on limits, depreciation, endorsements, exclusions, and settlement arrangements. A handler must confirm these before approving the amount.

Business and functional requirements

ID	Requirement	Priority
BR01	Give claims staff a complete view of outstanding work and its owner.	Must
BR02	Prevent release of payments without coverage verification and authority.	Must
BR03	Report claim delays using consistent event timestamps.	Must
FR01	Create a unique claim and retain the intake channel and source reference.	Must
FR02	Match policy and coverage to the loss timestamp, including effective amendments.	Must
FR03	Determine required documents by product, loss cause, and claim stage.	Must
FR04	Create deduplicated exceptions with rule version, owner, severity, and due time.	Must
FR05	Assign and reassign assessors with reason and assignment history.	Must
FR06	Record versioned settlement approval and validate cumulative payment commitments.	Must
FR07	Retain every status transition, closure, reopening, and associated reason.	Must
FR08	Provide dashboard filters and claim drill through with access restrictions.	Must
FR09	Reconcile incoming records and financial totals for each batch.	Must
FR10	Send due and overdue notifications without repeating the same notification.	Should

Nonfunctional requirements

NFR01: In the proposed workload test, the exception list should load within three seconds at the 95th percentile with 50 concurrent users and 50,000 open claims. NFR02: Operations data should be available within 15 minutes after a successful source event; the daily finance reconciliation completes by 07:00 UTC. NFR03: Access is role based, with sensitive fields masked in reporting. NFR04: Approval, rule, and record changes must include actor, timestamp, reason, and previous value. NFR05: A failed load must restart without duplicate claims, payments, or alerts. These are acceptance targets, not achieved performance.

User stories and acceptance criteria

US01 Resolve a policy mismatch

As a claims handler, I want unmatched policy references to enter a review queue so that I can correct the record without losing the loss report.

Given a claim with no matching policy version, when verification runs, then one active E01 exception is created, coverage stays unverified, and payment release is blocked. When a valid correction is accepted, the old reference remains in the audit history and verification runs again.

US02 Collect the right documents

As a claims handler, I want a product-specific document checklist so that I can see what is still needed before assessment.

Given a collision claim requiring photos and a repair estimate, when only photos are accepted, then the estimate task is shown as pending. Uploading a file alone does not complete the task. Only an accepted document of the required type closes it. Rejected or superseded files remain in history.

US03 Control settlement payments

As a finance approver, I want new requests checked against remaining approval so that several payments cannot exceed the amount authorized.

Given approved authority of USD 3,500 and USD 2,000 committed, when a USD 1,800 request is submitted, then it is blocked. A USD 1,500 request may pass if coverage is verified, no holds exist, the approval is current, and required roles approve it. Two concurrent requests must share the same locked balance check.

US04 Track a reopened claim

As a claims manager, I want reopened claims shown separately so that I can distinguish first closure from subsequent work.

Given a closed claim, when an authorized handler reopens it, then a reason and event timestamp are required, the previous closure remains in history, and current open ageing continues from the original FNOL. Reopen episode ageing starts at the latest reopening.

US05 Explain a dashboard number

As a manager, I want to open the claims behind a KPI so that I can check its denominator and take action.

Given a selected snapshot date and queue, when I select the SLA breach count, then only eligible claims with a breached active assessment task appear. The screen shows snapshot time, rule version, task age, owner, and exclusions. Unauthorized users cannot open other queues through exports or direct links.

Exception rules and routing

Rule	Trigger	Action and owner
E01	No unique policy version matches the reported reference and loss time.	High. Coverage queue. Payment hold.
E02	Loss outside the selected coverage period or unresolved coverage terms.	High. Coverage specialist review. No automatic denial.
E03	Required document is absent or not accepted at the relevant stage.	Medium. Handler follows up and records customer dependency.
E04	Same policy, vehicle, loss date, and cause suggest another claim.	High. Investigation queue. Candidate match only.
E05	Active assessment task age exceeds 48 elapsed hours.	Medium. Assessment supervisor reallocates or explains delay.
E06	Proposed payment plus existing commitments exceeds active approved authority.	Critical. Block request. Finance review.
E07	Source record count or signed financial total differs from loaded total.	High. Data operations. Withhold affected reporting partition.
E08	Closed claim has an unresolved payment hold or active required task.	High. Claims manager reviews attempted closure.

Service levels and priority

Proposed targets are 4 hours to acknowledge critical issues, 8 hours for high issues, and 24 hours for medium issues.

Assessment task age has its own 48-hour rule. These are internal scenario targets, not legal deadlines. Keep customer waiting time visible even where an approved internal timer pause exists. Any pause needs a reason, start, end, and authorizer.

Exception lifecycle

New → Assigned → In review → Resolved. A waived issue requires authority, reason, and expiry. Resolution requires fresh validation; manually changing the status is insufficient. Use claim ID, exposure ID where relevant, rule ID, and source object as the deduplication key. Reopen an existing issue if the condition returns; retain all episodes.

Data model and reporting grain

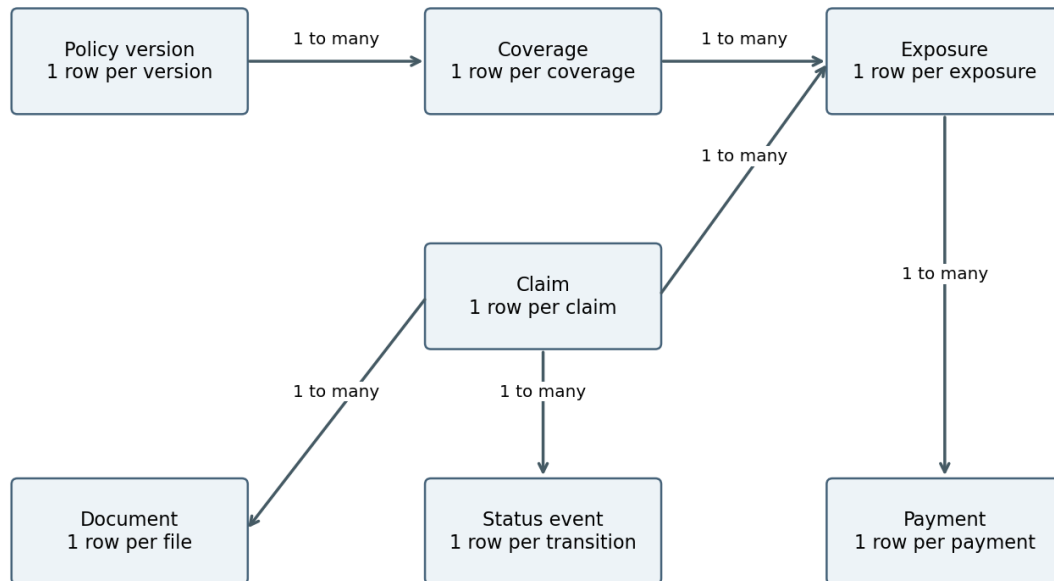


Figure 2 Simplified core relationships. The claim links to the verified policy version; this link is omitted from the picture to keep it readable.

A claim may have multiple exposures. Payments must join through exposure, while claim documents and events are separate child tables. Joining all child tables directly would multiply records and inflate payment totals. Aggregate each child table to the intended grain before creating a claim summary.

Entity	Key and essential fields
PolicyVersion	policy_version_id; policy_number; valid_from; valid_to; recorded_at; status; vehicle_id. Effective ranges are half open.
Coverage	coverage_id; policy_version_id; coverage_type; effective_from; effective_to; limit_amount; deductible_amount; currency.
Claim	claim_id; source_claim_ref; reported_policy_number; policy_version_id; fnol_at; loss_at; current_status; current_queue.
Exposure	exposure_id; claim_id; coverage_id; exposure_type; currency; active_approval_id.
Approval	approval_id; exposure_id; version; approved_amount; approved_at; approved_by; current_flag.
Payment	payment_id; exposure_id; signed_amount; status; external_reference; requested_at; posted_at; idempotency_key.
Document and StatusEvent	document_id and event_id respectively; claim_id; type or status; event time; actor; validation or reason.
Exception and Task	exception_id or task_id; claim_id; exposure_id; rule; owner; opened_at; due_at; closed_at; status.

Source to target mapping

The mapping below is a worked sample rather than a complete production interface contract. Raw staging retains the original values. Rejected values must not be silently replaced with plausible defaults. UTC and currency rules are agreed before loading.

Source field	Target field	Transformation and validation
claim_feed.claim_no	Claim.source_claim_ref	Trim spaces; preserve source system; unique within that system. Assign internal claim_id.
claim_feed.policy_no	Claim.reported_policy_number	Trim and uppercase for matching; keep raw reference for audit.
claim_feed.loss_datetime	Claim.loss_at	Parse declared source timezone to UTC; reject invalid or future loss timestamp for review.
claim_feed.notified_datetime	Claim.fnol_at	Convert to UTC; FNOL before loss requires review, not silent date repair.
policy_feed.effective_from and to	PolicyVersion.valid_from and valid_to	Create versioned half-open ranges. Reject overlaps for the same policy where uniqueness is required.
coverage_feed.type_code	Coverage.coverage_type	Lookup COLL → Collision and COMP → Comprehensive. Unknown codes enter quarantine.
document_feed.review_status	Document.validation_status	Map accepted, rejected, pending, and superseded explicitly.
approval_feed.amount	Approval.approved_amount	Decimal with 2 places; nonnegative; preserve approval version and currency.
payment_feed.amount and direction	Payment.signed_amount	Debit positive; reversal negative; reversal references original. Reject unknown direction.
payment_feed.status	Payment.status	Map requested, authorized, posted, voided, and reversed. Only eligible statuses enter the specified measure.
event_feed.status and time	StatusEvent.status and event_at	Map approved codes, retain event sequence and event ID. Deduplicate by source event key.
batch metadata	All staging records	Retain batch_id, ingested_at, source_file, source_record_id, and rule_version.

Mapping decisions to confirm

Ask source owners how cancellations and reinstatements are represented, whether payment reversals arrive as signed entries, how corrections are replayed, and which timestamp marks business completion. Keep these questions in a decision log until resolved.

Data lineage and integration controls

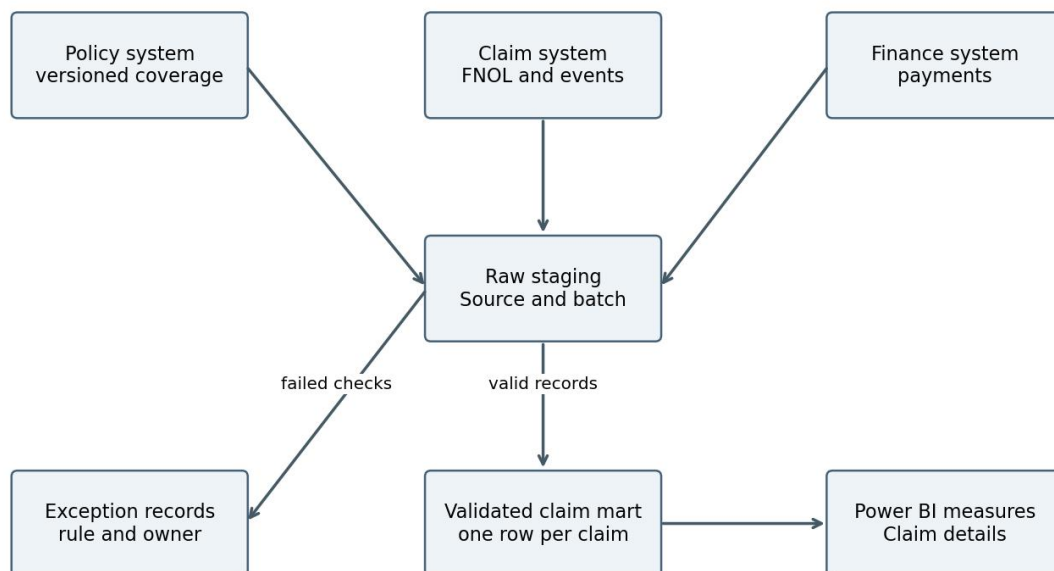


Figure 3 Proposed source to reporting flow. Exceptions are retained alongside usable records; financial failures block only the affected reporting partition.

One measure traced to its source

Assessment breach rate starts with `task_feed.assigned_at` and `completed_at`. STTM converts these to UTC and maps the task type. The validated task table identifies active assessment tasks at the snapshot. SQL compares their age with the 48-hour threshold. Power BI divides breached eligible claims by all eligible claims. UAT T06 checks the exact boundary and T12 checks the drill-through denominator.

Reliable loads

Use source event IDs or idempotency keys for deduplication. Retain load watermarks only after a successful commit. Send malformed records to quarantine with a reason and replay them after correction. Reprocess late events into affected snapshots without rewriting the original source history. Reconcile counts and signed financial totals by source, batch, currency, and status.

Payment control placement

A reporting query can find an overpayment after the event; it cannot safely prevent one. The blocking check belongs in the payment service inside a transaction, using a lock or equivalent concurrency control on the exposure balance. Include posted payments and authorized outstanding requests, exclude voided requests, and apply validated reversals once. The dashboard reports the control outcome.

Access and retention

Handlers see assigned claims; supervisors see their teams; finance sees payment fields. Keep bank details and personal identifiers outside the portfolio dataset. In production, retention and deletion periods require the insurer's approved policy. Log exports and approval changes. Use a separate service identity for ingestion.

SQL validation examples

The following examples use SQLite syntax so they can be checked locally. Production SQL must be adapted to the chosen platform. Claims and exposures use the grains described earlier. The code is read-only and identifies review candidates.

Q01 Unmatched policy or loss outside coverage period

```
SELECT c.claim_id,  
  CASE WHEN p.policy_version_id IS NULL THEN 'E01'  
        WHEN c.loss_at < p.valid_from  
              OR c.loss_at >= p.valid_to THEN 'E02' END AS rule_id  
FROM claims c  
LEFT JOIN policy_versions p  
  ON p.policy_version_id = c.policy_version_id  
WHERE p.policy_version_id IS NULL  
      OR c.loss_at < p.valid_from OR c.loss_at >= p.valid_to;
```

This checks the linked version's dates only. Production coverage verification also checks coverage effective dates, endorsements, vehicle eligibility, cancellations, and exclusions.

Q02 Potential duplicate claims

```
SELECT a.claim_id, b.claim_id AS candidate_claim_id  
FROM claims a JOIN claims b  
  ON a.claim_id < b.claim_id  
  AND a.policy_version_id = b.policy_version_id  
  AND a.vehicle_id = b.vehicle_id  
  AND date(a.loss_at) = date(b.loss_at)  
  AND a.loss_cause = b.loss_cause;
```

This is a narrow matching rule for the exercise. Different exposures from one accident may be legitimate. The investigator checks evidence before merging or closing a record.

Q03 Posted payments above approved authority

```
WITH paid AS (  
  SELECT exposure_id, SUM(signed_amount) AS net_paid  
  FROM payments WHERE status = 'POSTED'  
  GROUP BY exposure_id  
)  
SELECT e.exposure_id, e.approved_amount, p.net_paid  
FROM exposures e JOIN paid p USING (exposure_id)  
WHERE p.net_paid > e.approved_amount;
```

The SQL fixture stores the current approved amount on exposures for simplicity. The proposed production model obtains it from the active approval version. This query detects an existing breach; request prevention also includes pending authorized commitments.

SQL results and control verification

Q04 Assessment tasks beyond the agreed threshold

```
SELECT task_id, claim_id,
       ROUND((julianday('2026-10-01T00:00:00')
              - julianday(assigned_at)) * 24, 2) AS age_hours
FROM tasks
WHERE task_type = 'ASSESSMENT' AND status = 'OPEN'
      AND (julianday('2026-10-01T00:00:00')
            - julianday(assigned_at)) * 24 > 48;
```

Synthetic test fixture

A small fixture uses policy PV1 valid from 1 January 2026 to 1 January 2027. C1 and C2 share the same vehicle, loss date, and collision cause. C3 has no policy match. C4 has a loss before the selected policy starts. X1 has USD 3,500 approved and posted entries of USD 2,000 and USD 1,800. X2 has USD 3,000 approved, a USD 1,000 debit and a USD 200 reversal. T1 was assigned 72 hours before the snapshot; T2 exactly 48 hours before it.

Query	Expected output	Checked outcome
Q01	C3 → E01; C4 → E02	Matched
Q02	C1 paired with C2 only	Matched
Q03	X1 → net paid 3,800 above approval 3,500; X2 excluded	Matched
Q04	T1 → 72 hours; T2 excluded at exact 48-hour boundary	Matched

These results validate the four example queries against designed edge cases. They do not estimate an insurer's real exception rates or demonstrate production readiness. Missing-document checks, request blocking, role restrictions, and concurrency require the UAT scenarios and application tests below.

Data quality checks before reporting

Check uniqueness at every grain, required IDs and timestamps, source code lookup coverage, policy-version overlaps, child records without parents, unknown currencies, and events arriving out of order. Reconcile debit and reversal totals independently before computing the net. Publish a freshness indicator and a rejected-record count with every refresh.

Dashboard specification and metric definitions

The proposed Power BI report has three pages. It is a design specification; no live Power BI file is claimed. Use a date dimension and shared claim, queue, coverage, and channel dimensions. Keep payments, task snapshots, and claim summaries as separate facts with deliberate relationships.

Page	Views and decisions
Operations overview	Cards for open claims, overdue assessment claims, unassigned claims, and open exceptions. Age buckets and queue workload identify where to act.
Delay analysis	Median and 90th percentile time to first settlement by intake cohort. Stage duration and document waiting time help explain delays.
Payment controls	Blocked requests, posted overpayments, reconciliation breaks, and claim-level audit details. Restrict finance drill through.

Metric	Definition and denominator
Open claim ageing	Snapshot minus FNOL for claims open at snapshot. Reopened claims retain original ageing; show episode ageing separately.
First settlement time	First valid settlement timestamp minus FNOL, for claims first settled in the selected cohort. Exclude missing or invalid dates and disclose excluded count.
Assessment breach rate	$\text{Distinct claims with an active assessment task older than 48 hours} \div \text{distinct claims with an active assessment task at snapshot}$.
Document completeness	$\text{Claims with every applicable required document accepted} \div \text{claims whose stage requires those documents at snapshot}$.
Payment request block rate	$\text{Distinct submitted requests blocked by a control} \div \text{distinct requests evaluated in period}$. Separate request IDs from retry attempts.
Reopen rate	$\text{Claims first closed in a cohort and reopened within 30 days} \div \text{that cohort's claims with a complete 30-day follow-up}$.
Net paid and recovery	Posted signed claim-cost payments and posted recoveries shown separately. Do not subtract recoveries twice or mix expense categories.

Display snapshot time, currency, cohort basis, and active filters. Selecting a card opens the exact underlying records. Do not calculate loss ratio from this claims-only dataset: earned premium and agreed expense treatment are missing.

UAT scenarios and traceability

These are planned acceptance scenarios. Apart from the four SQL fixture checks, they have not been executed against an application. The business owner signs off only after evidence is recorded.

Test	Scenario and expected result	Links
T01	Invalid policy reference creates one E01 and blocks payment; correction retains history.	FR02 / US01 / E01
T02	Loss exactly at coverage end enters review; effective start is eligible for date checking.	FR02 / E02
T03	An endorsement effective after the loss does not overwrite the applicable version.	FR02 / E02
T04	Uploaded but rejected estimate remains outstanding; accepted replacement completes task.	FR03 / US02 / E03
T05	Candidate duplicate is flagged without automatic fraud or denial decision.	FR04 / E04
T06	48-hour assessment task passes boundary; 48 hours plus 1 second breaches.	FR04 / E05
T07	2,000 committed plus 1,800 request exceeds 3,500 approval and is blocked.	FR06 / US03 / E06
T08	Two simultaneous 1,000 requests against 1,500 remaining allow at most one.	FR06 / NFR05
T09	Validated reversal reduces net paid once; repeated delivery has no additional effect.	FR06 / FR09
T10	Closure with active payment hold fails; reopening requires reason and preserves closure.	FR07 / US04 / E08
T11	Missing source records prevent publication of the affected finance partition.	FR09 / E07
T12	Breach card count equals distinct claims in drill through under the same filters.	FR08 / US05
T13	Handler cannot view another queue through direct link or exported file.	NFR03 / FR08
T14	Failed load replay creates no duplicate rows or notifications.	NFR05 / FR09
T15	Late closure event repairs the historical snapshot and records the correction.	FR07 / FR09

For every execution, record build version, fixture, expected result, actual result, evidence link, tester, date, and defect ID. Add tests for assignment and reassignment, alert retry suppression, approval version changes, performance, and audit-field completeness before release.

Delivery plan risks and benefit measurement

Phase	Work and exit condition
Week 1	Confirm scope, terminology, source owners, and coverage assumptions. Agree the business case and process.
Week 2	Complete requirements, data dictionary, STTM, rule catalogue, and metric definitions. Resolve open mapping decisions.
Week 3	Build the synthetic model and ingestion prototype. Verify exceptions and reconciliation.
Week 4	Build reporting views and dashboard prototype. Review denominator and drill-through consistency.
Week 5	Execute application UAT, concurrency tests, role tests, and load replay. Resolve critical defects.
Week 6	Run a limited pilot, train handlers, monitor queues, and decide whether to expand.

Risks and responses

Incorrect coverage rules could block valid claims. Keep coverage decisions with specialists and version the rules. Poor timestamps could distort ageing. Preserve source timezone and quarantine invalid dates. Excess alerts could hide urgent work. Deduplicate and monitor alert volume. Child-table joins could inflate money totals. Aggregate at exposure or claim grain and reconcile. Unclear ownership could leave exceptions untouched. Agree fallback queues and escalation before the pilot.

How benefits would be measured

Use four weeks of baseline and four weeks of pilot data for comparable coverage and claim-complexity groups. Track intake cohorts and unresolved claims alongside completed claims to avoid making faster closures look like universal improvement. Compare median and 90th percentile handling times, touches per claim, assessment breaches, and payment control outcomes. Explain changes in workload, staffing, and catastrophe volume. An observational comparison alone cannot prove causation.

Illustrative targets: reduce median first settlement time by 15%, lower repeated document requests by 20%, and achieve 100% named ownership for active exceptions. These targets require baseline confirmation. No achieved percentages are claimed.

A simple financial estimate

Assume 1,000 claims a month, 10 minutes of manual checking saved per claim, and USD 30 per staff hour. Potential gross capacity value is $1,000 \times 10 \div 60 \times 30 = \text{USD } 5,000$ per month. This is a scenario calculation, not cash savings. Deduct build and running costs, and confirm whether freed capacity reduces cost or supports more work.

Recruiter walkthrough and interview preparation

A short project explanation

“I designed a P&C motor claims case study around a practical problem: claims get delayed when information is incomplete and nobody owns the next action. I mapped the journey from loss reporting to payment, wrote requirements and acceptance criteria, and defined exception rules. I also mapped the source fields, checked sample SQL against synthetic cases, and specified a dashboard. The most useful control was checking cumulative payments against the approved amount, including outstanding commitments.”

Why did you choose this problem

“It let me connect the customer journey with operations and data. A dashboard can show a delay, but a handler also needs to know what caused it, who owns it, and what would resolve it.”

How did you understand the insurance domain

“I used public insurance and claims-system documentation to understand FNOL, exposures, coverage, reserves, and payments. I kept policy assumptions explicit. I would validate the detailed rules with claims and coverage specialists before implementation.”

What was the hardest technical issue

“A claim can have several documents, events, and payments. Joining those tables directly can repeat payments and inflate totals. I defined the grain first and aggregated each child table before building claim-level reporting.”

How would you prove improvement

“I would compare a documented baseline with a limited pilot, using similar claim groups. I would check slow cases and unresolved cases as well as completed claims. The document contains proposed targets, not results from a live insurer.”

Suggested portfolio entry

Motor Claims Delay and Payment Control | Independent P&C case study. Designed claim workflows, requirements, source to target mapping, exception rules, SQL validation examples, dashboard specifications, and UAT scenarios using synthetic examples.

What to show during a five minute walkthrough

Start with the worked claim. Show the process diagram, payment rule, source mapping, query result, and related acceptance test. Finish with the dashboard decision it supports. Describe the work as a portfolio design and SQL exercise; do not imply employment or production delivery at an insurer.

References glossary and open decisions

Public references were checked on 5 October 2026. They support terminology and lifecycle concepts, not the fictional business problem, proposed SLAs, benefits, or implementation results.

1. Insurance Information Institute. Auto Insurance Basics. <https://www.iii.org/article/auto-insurance-basics>
2. Insurance Information Institute. What is covered by collision and comprehensive auto insurance. <https://www.iii.org/article/what-is-covered-by-collision-and-comprehensive-auto-insurance>
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Document abbreviations

BRD: Business requirements document. FR: Functional requirement. STTM: Source to target mapping. RTM: Requirements traceability matrix. UAT: User acceptance testing. SLA: Service level agreement. FNOL: First notice of loss. BI: Business intelligence. NFR: Nonfunctional requirement.

Decisions needed before implementation

Decision	Proposed owner
Which product, market, and policy wording apply	Product and coverage owner
What makes a document required or accepted	Claims operations
Which statuses count as payment commitments	Finance and payment service owner
How approval amendments affect pending requests	Finance approver
Whether service timers pause and under what conditions	Claims manager
Which system is authoritative for each event	Data owners
What retention, recovery, and access requirements apply	Security and business owners

Version 1.0. Prepared as an independent portfolio case study. The next practical step is to build the proposed application controls and Power BI report, then record actual UAT evidence before changing the project status.